Conditional Statement

A conditional statement in Python is used to make decisions based on whether a condition is True or False.

TYPES:

if Checks a condition

elif Checks another condition

else Executes when previous conditions are false

Nested if An if inside another if

1.if

In [6]:

```
num = 20
if num > 0:
  print("The number is positive")
```

```
The number is positive
```

2.else

In [9]:

```
a = 20
b = 30
if b > a:
  print("a is greater")
else:
  print("b is greater")
```

```
a is greater
```

3.elif

In [12]:

```
a = 22
b = 33
if b > a:
  print("b is greater than a")
elif a == b:
  print("a and b are equal")
else:
  print("a is greater than b")
```

```
b is greater than a
```

4.Nested_if

In [15]:

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

```
x = 41
if x > 10:
  print("Above ten,")
  if x > 20:
    print("and also above 20!")
  else:
    print("but not above 20.")
```

```
Above ten,
and also above 20!
```

Practice Question

1. Write a Python program to check whether a number is positive, negative, or zero.

In [25]:

```
num = int(input("enter num:"))
if num > 0:
    print("positive")
elif num <0:
    print("negative")
else:
    print("zero")
```

```
negative
```

2. Write a Python program to check whether a number is even or odd.

In [28]:

```
num = int(input("enter num:"))
if num%2==0:
    print("even")
else:
    print("odd")
```

```
even
```

3. Write a Python program to find the largest of two numbers.

In [31]:

```
a=10
b=20
if a >b:
    print("a is great")
else:
    print("b is great")
```

```
b is great
```

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

4. Write a Python program to find the largest of three numbers.

In [34]:

```
a=10
b=4
c=1
if a > b and a>c:
    print("a is great")
elif b>a and b>c:
    print("b is great")
else:
    print("c is great")
```

```
a is great
```

5. Write a Python program to check whether a person is eligible to vote based on their age.

In [37]:

```
age = int(input("enter age:"))
if age > 18:
    print("eligible to vote")
else:
    print("not eligible")
```

```
not eligible
```

6. Write a Python program to check whether a year is a leap year.

In [46]:

```
year = int(input("enter year:"))
if year%4==0 and year%400==0:
    print("leap")
else:
    print("not leap")
```

```
not leap
```

7. Write a Python program to calculate a student's grade based on their marks.

- 90 and above: A Grade

- 75–89: B Grade

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

- 50–74: C Grade

- Below 50: Fail

In [50]:

```
mark = int(input("enter mark:"))
if mark >= 90:
    print("Grade A")
elif mark>=75 and mark<=89:
    print("Grade B")
elif mark>=50 and mark<=74:
    print("Grade C")
else:
    print("Fail")
```

```
Grade C
```

8. Write a Python program to check whether a number is divisible by both 5 and 11.

In [57]:

```
num = int(input("enter num:"))
if num%5==0 and num%11==0:
    print("Yes")
else:
    print("No")
```

```
Yes
```

9. Write a Python program to check whether a person is eligible for a job based on age and qualification using a nested if statement.

In [62]:

```
age = int(input("enter age:"))
qua = input("enter qualification:")
if age >=22:
    if qua =="BE":
        print("eligible for job")
    else:
        print("not eligible")
else:
    print("invalid age")
```

```
eligible for job
```

10. Write a Python program to calculate an electricity bill based on the number of units consumed using if-elif-else.

In [65]:

```
unit = int(input("Enter units: "))
if unit <= 100:
```

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

```
    bill = unit * 2
elif unit <= 200:
    bill = unit * 3
else:
    bill = unit * 5

print("bill:", bill)
```

```
bill: 100
```

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF